

Background

- ❖ **Dead Zones** are areas of water with little to no oxygen. Dead zones typically form in deeper waters and imbalance marine ecosystems [4]
- ❖ Habitats in the **Chesapeake Bay** are susceptible to pollutants and low oxygen levels
 - The **largest dead zone** in the Chesapeake Bay spans **1.2 cubic miles**

Inspiration

We drew inspiration from the **green sea turtle** for their **efficient underwater mobility** and **hydrodynamic shells** [1]



- ❖ Sea turtles can **alter** their **buoyancy** in the water allowing them to dive and rise with ease [2]
- ❖ Sea turtles use their **long front flippers** to accelerate [3]

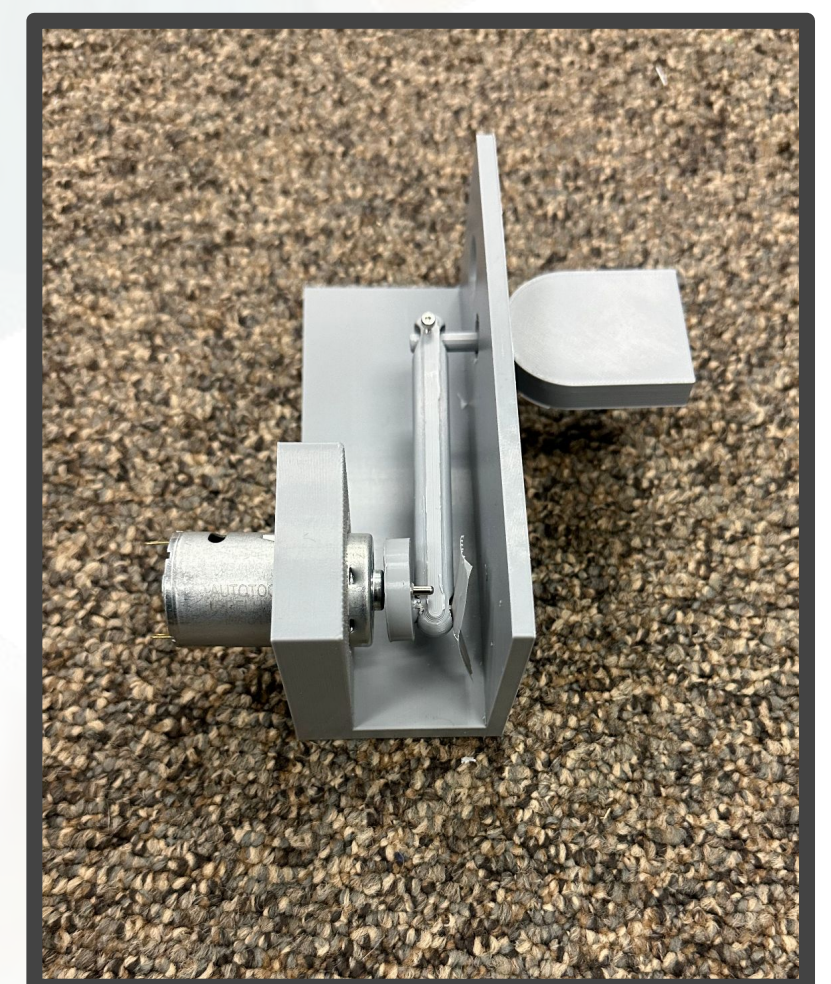
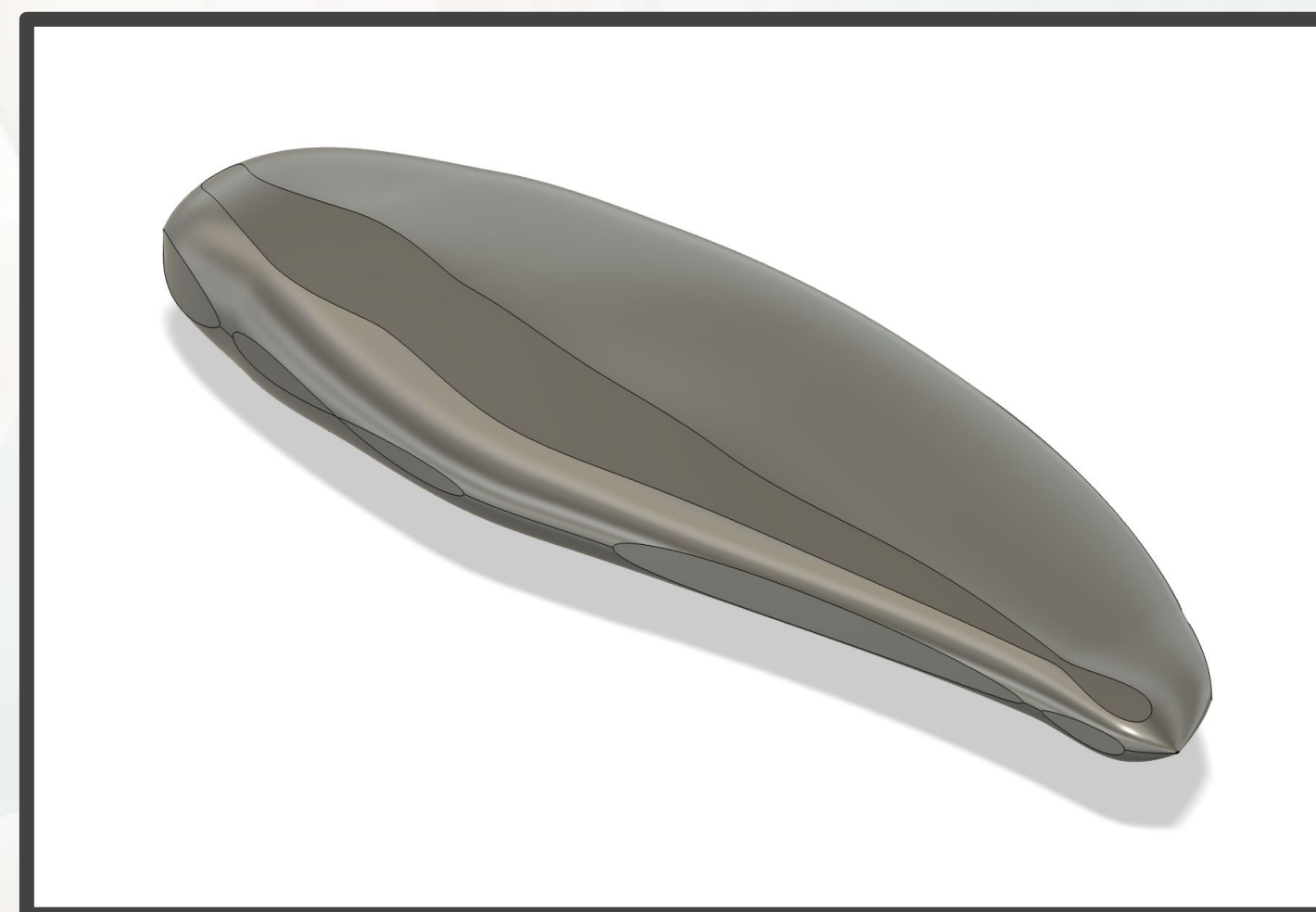
Project Goals

- ❖ Build a robot with **sensors** able to **assess water health** and diagnose issues
- ❖ The robot must endure a wide range of **temperatures** and variable **water pressure**
- ❖ Mimic sea turtle's **swimming mechanisms** and hydrodynamic shape
- ❖ The robot will also interact with **marine life** like fish and crabs

Project Design

Sea Terp features the following:

- ❖ Arduino-compatible **oxygen sensor** to monitor water quality coupled with an interior **canal** for water flow
- ❖ **Camera** to collect visual **data samples** from dead zones and other habitats
- ❖ **Buoyancy chamber** to ascend and descend underwater
- ❖ **Bio-inspired flippers** for steering
- ❖ **Bio-inspired shell** to house sensors and provide hydrodynamic shape
- ❖ Rear-mounted **propellers** to provide thrust



Future Steps

- ❖ Conduct **field trials** of the robot in the **Chesapeake Bay** to validate its performance in real-world conditions
- ❖ Program a **remote controller** to allow for Sea Terp to be manually controlled.
- ❖ Design a better **crankshaft system** to allow for a **greater range of motion** for the flippers.

References

[1] H. Brown, "What you need to know about Hawaiian green sea turtles," Sea Paradise, <https://www.seaparadise.com/what-you-need-to-know-about-hawaiian-green-sea-turtles/> (accessed Apr. 29, 2024).

[2] N. van der Geest, L. Garcia, R. Nates, and D. A. Godoy, "New insight into the swimming kinematics of wild green sea turtles (*Chelonia mydas*)," *Scientific Reports*, vol. 12, no. 1, Oct. 2022. doi:10.1038/s41598-022-21459-y

[3] Larisa Bennett, Reviewed by Connie Y. Kot (Duke University Marine Laboratory), "Sea turtles," Smithsonian Ocean, <https://ocean.si.edu/ocean-life/reptiles/sea-turtles#:~:text=To%20help%20theme%20%20efficiently%20power,rudders%20to%20help%20theme%20%20steer.> (accessed Apr. 29, 2024).

[4] "Dead Zones," www.cbf.org, <https://www.cbf.org/issues/dead-zones/index.html#:~:text=What%20Causes%20Dead%20Zones%20in> (accessed Apr. 30, 2024).